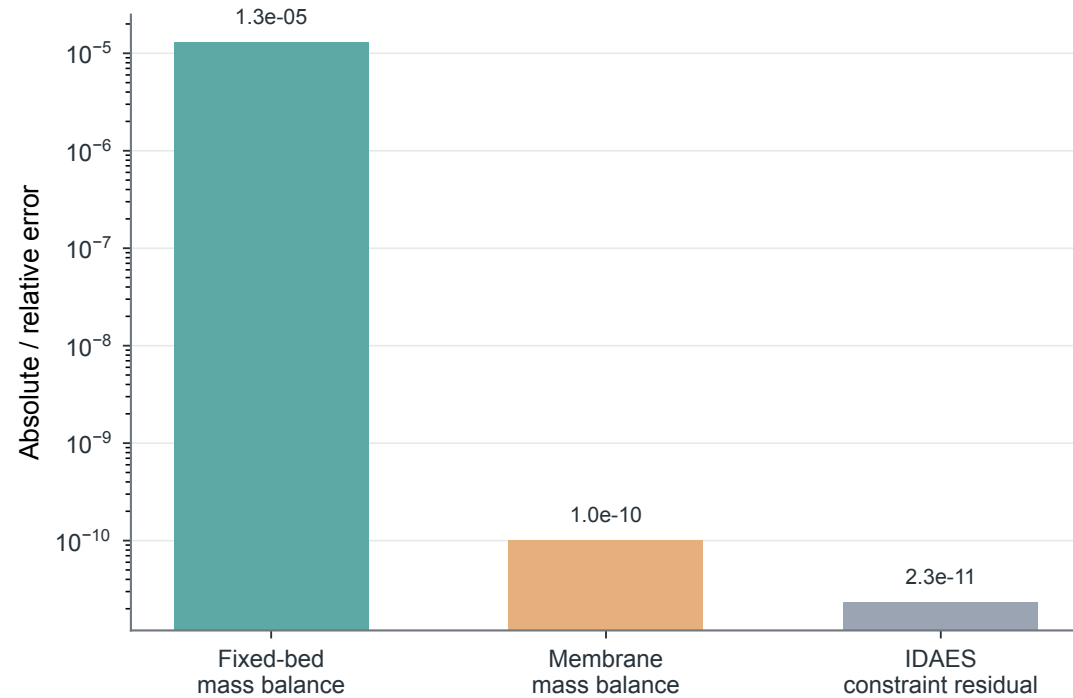
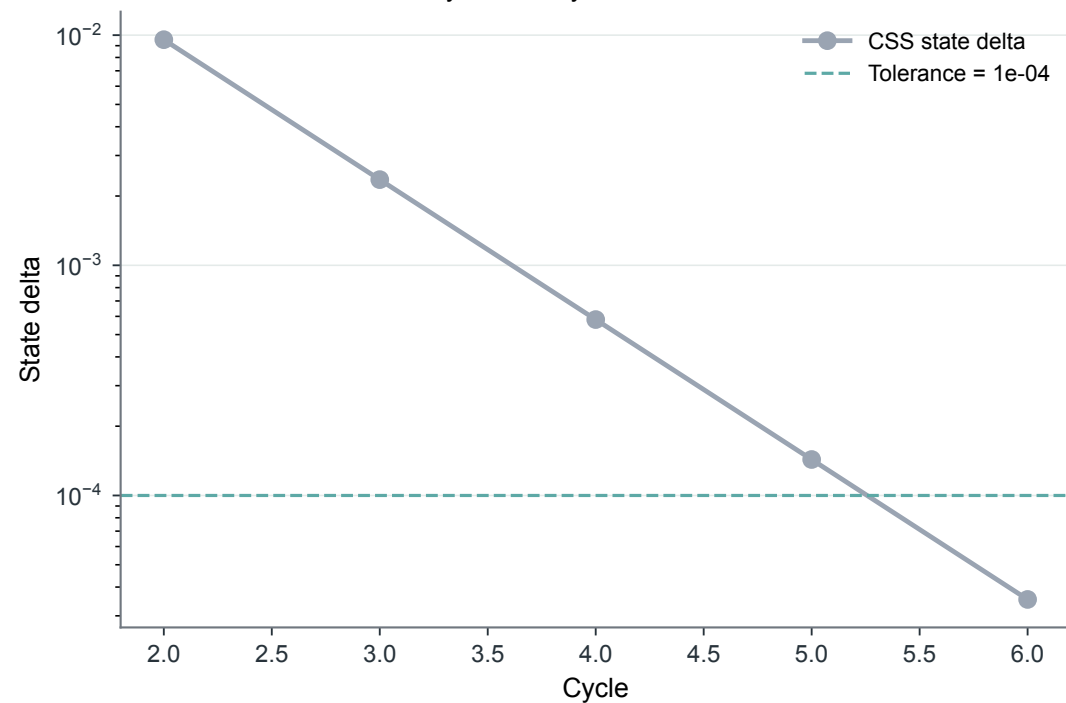


Numerical validation and reasonableness evidence

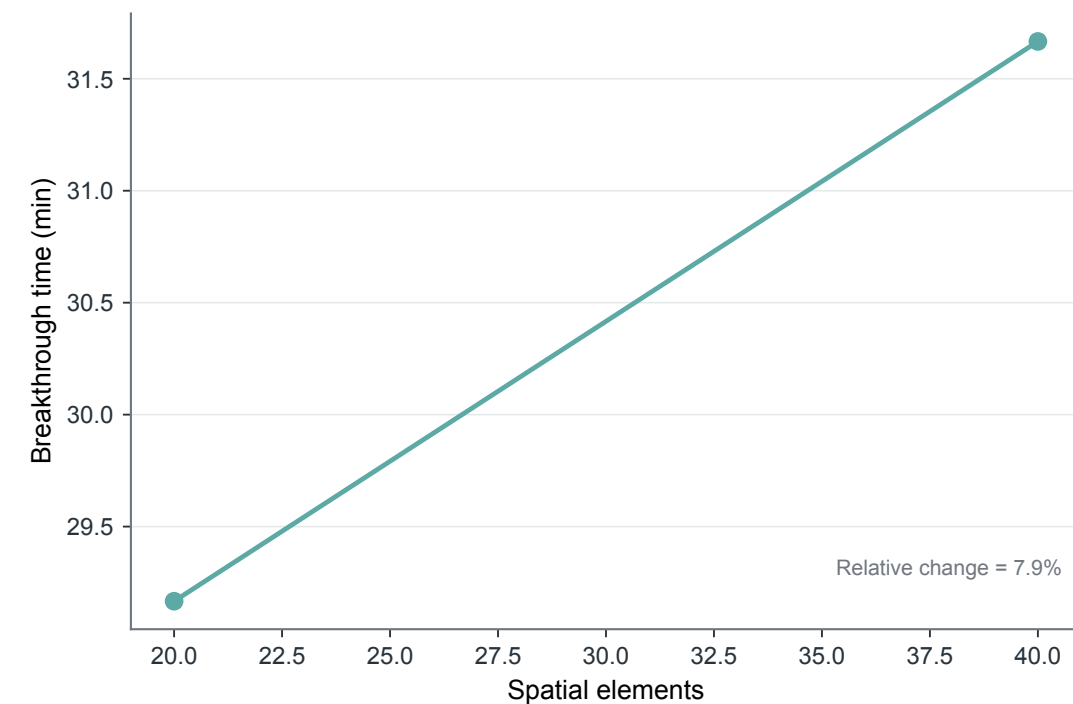
Balance and residual checks



Cyclic steady-state evidence



Fixed-bed mesh check



Validation layers

1. Code: 14 pytest tests passed.
2. Units / bounds: ModelSpec preflight passed.
3. Numerical: ODE/DAE termination successful.
4. Conservation: errors reported at left.
5. CSS: final delta below tolerance.
6. Physical: monotonic $q^*(P)$, positive rates, non-negative flows and closure checks.

Not included: experiment, literature benchmark, full distributed PSA, or uncertainty quantification.